

Jaeyong Ko

M.S. Student
Industrial Engineering
Seoul National University
Data Science & Business Analytics Lab

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RESEARCH INTERESTS

- **Current Focus:** Visual Reasoning, Interleaved Vision-Language Reasoning, Latent Reasoning
- **Previous Research:** LLM Mathematical Reasoning, LLM Evaluation in the Financial Domain

EDUCATION

- **M.S.** *Industrial Engineering* Sep. 2025 – Present
Seoul National University *Advisor: Prof. Pilsung Kang*
- **B.S.** *Industrial Engineering* Mar. 2019 – Aug. 2025
Seoul National University *GPA: 3.72/4.0*

PUBLICATIONS

FinLifeBench: Exhaustive Life-Event History and Financial-State Reconstruction from Longitudinal Banking Dialogue 2026

Hangyeul Lee*, Juyoung Oh*, Jaeyong Ko, Sunmin Kim, Jaeik Park, Hyunkyu Kim, Jungmin Son, Pilsung Kang[†].
ACM ICAIF 2026 (Accepted). [arXiv]

Cliff Tokens: Analyzing Failure Trigger Tokens in LLM Mathematical Reasoning 2026

Jaeyong Ko, Jinu Lee, Pilsung Kang, Yukyung Lee[†].
arXiv preprint arXiv:2606.25524. [project page] [arXiv] [github] [dataset]

EDUCATIONAL CONTRIBUTIONS

Can a model think VISUALLY? 2026

Jaeyong Ko, Eunseo Kim, Doehyeon Lee.
NeurIPS 2026 Education Track (Accepted).

AI Safety Lives in a Direction: Reading, Breaking, and Repairing Refusal in Multimodal Models 2026

Doehyeon Lee, Yoo-Min Jung, Jaeyong Ko.
NeurIPS 2026 Education Track (Accepted).

RESEARCH EXPERIENCE

- **Research Lead** *Visual Latent Reasoning for Unified Models* Jul. 2026 – Present
Transferring visual reasoning capabilities from the generation branch of unified models to the understanding branch
- **Research Intern** *École Polytechnique de Montréal* Apr. 2024 – Aug. 2024
Human-centric Smart Manufacturing Workcells for Industry 5.0 applications. *Advisor: Prof. Walid Jomaa*
- **Research Intern** *Financial Engineering Lab, KAIST* Dec. 2023 – Feb. 2024
Reinforcement learning applications for goal-based wealth management. *Advisor: Prof. Woo Chang Kim*

PROJECTS

- **Development of Core Foundation Model and AI Agent Technology for Thermal-Process Manufacturing** | *Granted by Ministry of Trade, Industry and Energy* Sep. 2026 – Present
 - Designed the backbone architecture for the thermal-process foundation model and established an integrated learning framework utilizing domain data and PINN.
- **Duo 2.0: Evaluating LLM Reliability in Long Financial Conversations** | *Granted by KakaoBank* Dec. 2025 – Present
 - Quantified LLM capabilities in resolving ambiguous user queries using LLM-as-a-Judge methodologies.
 - Designed a financial life-cycle event conversation benchmark for domain-specific conversational reliability.
- **Development of Advanced Integrated Monitoring Technologies for Laser Welding** Jul. 2025 – Dec. 2025
| *Granted by LG Electronics*
 - Designed and conducted experiments for a laser welding anomaly detection model utilizing time-series and image multimodal data.

INDUSTRY EXPERIENCE

- **President & Team Member** *SNU Startup Consulting Group (ACT)* Mar. 2023 – Dec. 2023
 - Led a student consulting group and secured four startup consulting engagements with *Moderntec*, *Designovel*, *Spring Onward*, and *Sweetspot*.
 - Developed GTM strategies for *Designovel* (AI fashion-tech), *Moderntec* (EV charging infrastructure), and *Wavebridge* (cryptocurrency OTC platform).
- **Product Manager Intern** *EXGRAM* Jul. 2022 – Oct. 2022
 - Planned app configurations and push notifications for the app, Veloo.

TEACHING EXPERIENCE

- **Peer Tutor** *Production Control (406.310)* Sep. 2024 – Dec. 2024
Seoul National University
- **Peer Tutor** *Human Factors Engineering (406.304)* Sep. 2023 – Dec. 2023
Seoul National University

HONORS & AWARDS

- **Excellence Award** *Poster Presentation* Aug. 2026
2026 KDMS Summer Conference, Korea Data Mining Society

SKILLS

- **Programming and Software:** Python, R, MySQL, Docker, Git, $\LaTeX$.
- **Languages:** Native Korean; professional working proficiency in English (TOEFL 5.0/6.0, OPIc IH).

MILITARY SERVICE

- **Corporal** *Republic of Korea Army* Jan. 2021 – Jul. 2022
Completed full-term mandatory military service.